Civil Engineering

	Normative 4-year Degree
Major	Civil Engineering 土木工程
Degree / Award Title	Bachelor of Engineering in Civil Engineering 工學士(土木工程)
Offering Academic Unit	Department of Architecture and Civil Engineering
Normal Period of Study	4 years
Maximum Period of Study	8 years
Credit Units Required for Graduation	121 credit units

Note: The following curriculum information is subject to periodic review and changes.

GE Requirements (31 credit units) Catalogue Term : Semester A 2025/26

		Credit Units
GE University Requirements	GE1401 University English	3
	GE2410 English for Engineering	3
	GE1501 Chinese Civilisation - History and Philosophy	3
	GE1601 Whole-Person Development	1
GE Distributional Requirements	Take at least 1 course from each of the three distributional areas: Area 1: Arts and Humanities Area 2: Study of Societies, Social and Business Organisations Area 3: Science and Technology Visit the GE website for more course information.	12
College-specified Courses	MA1200 Calculus and Basic Linear Algebra I *Or* MA1300 Enhanced Calculus and Linear Algebra I	3

	MA1201 Calculus and Basic Linear Algebra II *Or* MA1301 Enhanced Calculus and Linear Algebra II	3
	CA2066 Professionals and Society	3

College / School Requirements (6 credit units) **Catalogue Term : Semester A 2025/26**

Course Code	Course Title	Credit Units
CS1302	Introduction to Computer Programming	3
PHY1201	General Physics I	3

Major Requirements (84 credit units) **Catalogue Term : Semester A 2026/27**

1. Core Courses (72 credit units)

Course Code	Course Title	Credit Units	Remarks
CA1167	Engineering Communication	3	
CA2560	Geology for Engineers	3	
CA2673	Engineering Mechanics	3	
CA2674	Construction Materials	3	
CA2675	Fluid Mechanics	3	
CA2676	Transportation Engineering	3	
CA3122	Engineering Analysis	3	
CA3144	Engineering Surveying	3	
CA3168	Building Information Modelling for Capital Projects	3	
CA3619	Design of Structural Elements	3	
CA3632	Mechanics of Structures and Materials	3	
CA3633	Structural Analysis	3	
CA3634	Reinforced and Prestressed Concrete Structures	3	
CA3677	Hydraulics and Hydrology	3	
CA3686	Construction Contract and Management	3	
CA3687	Soil Mechanics	3	
CA3704	Construction Engineering	3	
CA4708	Finite Element Methods in Civil Engineering	3	
CA4516	Final Year Project	6	A course to fulfil the internship/consultancy project/research project requirement for normative 4-year degree students

CA4526	Integrated Building Project Development (Civil Engineering)	3	
CA4665	Geotechnical Analysis and Design	3	
CA4682	Advanced Geotechnical and Foundation Engineering	3	

Select **ONE** from the following:

Course Code	Course Title	Credit Units	Remarks
CA2169	Environmental Engineering	3	
SEE2201	Fundamentals of Environmental Engineering	3	Pre-requisite of SEE2201 for ACE students: PHY1201 and MA1200 / MA1300 and MA1201 / MA1301

2. Electives (12 credit units)

Common Elective Course (3 credit units)

Select **ONE** from the following:

Course Code	Course Title	Credit Units	Remarks
CA2505	Industrial Training - Civil Engineering	3	Students who would like to join HKIE Scheme A training are advised to take this elective
CA2506	Industrial Internship	3	A course to fulfil the internship/consultancy project/research project requirement for normative 4-year degree students

Stream Elective Courses (9 credit units)

Students are required to choose one of the following streams to concentrate their study:

Structural Engineering Stream

Select **THREE** courses from the following:

Course Code	Course Title	Credit Units	Remarks
CA3401	Construction Management and Economics	3	
CA3691	Building Services for Building Professionals	3	
CA4423	Sustainable Green Construction	3	
CA4617	Temporary Works Design	3	
CA4644	Wind and Earthquake Engineering	3	
CA4679	Advanced Environmental Engineering	3	
CA4709	Design of Tall Buildings	3	

Infrastructure and Smart City Stream

Select **THREE** courses from the following:

Course Code	Course Title	Credit Units	Remarks
CA3342	Architecture, Smart Cities and Design	3	
CA3707	Modelling for Smart Digital Infrastructure System	3	
CA4190	Transport Policy and Planning	3	
CA4191	Transport Economics	3	
CA4192	Sustainable Urbanism	3	
CA4423	Sustainable Green Construction	3	
CA4424	Strategic BIM Management in Smart Construction	3	
CA4457	Infrastructure Asset Management	3	
CA4710	Highway Engineering	3	
CA4718	Power Electronics and Smart Lighting Controls	3	
CA4728	Smart Ventilation and Indoor Air Quality	3	

This Major is accredited by The Hong Kong Institution of Engineers (HKIE). Graduates are eligible for Scheme A Training in the following disciplines:

Programme	Discipline	Eligibility	Conditions (if applicable)
BEng Degree in Civil Engineering (with streams in Structural Engineering / Infrastructure and Smart City) (FT)	Civil (CVL)	Eligible	-
	Geotechnical (GEL)	Eligible	-
	Building (BUD)	Eligible subject to condition(s)	Graduates for this programme are eligible for Scheme "A" in Building (BUD) Discipline only if they have fulfilled the relevant top-up requirement as specified in the Admission Requirements for the Building Discipline.
BEng Degree in Civil Engineering (with stream in Structural Engineering) (FT)	Structural (STL)	Eligible subject to condition(s)	Graduates of this programme are eligible for Structural (STL) Scheme “A” Discipline on condition that they have to take any one of the following courses: (i) CA4644, (ii) CA4682, or (iii) CA4516, which must be in structural engineering.
BEng Degree in Civil Engineering (with stream in Infrastructure and Smart City) (FT)	Logistics & Transportation Engineering (LTE)	Eligible subject to condition(s)	This programme is eligible for Scheme “A” in Logistics & Transportation (LTE) Discipline on condition that the graduates must have taken CA2676 and CA3707.